Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project developed an end-to-end Sales Forecasting and Demand Intelligence System to support inventory planning and business decision-making. Historical sales data from the Superstore dataset was analyzed to identify sales trends, seasonal patterns, and future demand. Three forecasting models—SARIMA, Prophet, and XGBoost—were evaluated to predict future sales. XGBoost delivered the highest forecasting accuracy and was selected as the preferred model. Additional analyses, including anomaly detection and product demand segmentation, provided valuable insights into unusual sales patterns and product performance. A Streamlit dashboard was also developed to enable interactive business reporting.

Key Findings from Exploratory Data Analysis (EDA)

The exploratory data analysis identified several important business insights:

- Furniture generated the highest forecasted sales among the three product categories.
- Sales showed clear seasonal patterns, with stronger performance during certain months of the year.
- The average shipping time remained relatively consistent across regions.
- Monthly sales displayed an overall increasing trend with periodic fluctuations, indicating stable business growth.
- The West and East regions showed similar expected future demand levels.

The findings provide a strong foundation for demand forecasting and inventory planning.

Sales Forecasting Results

Three forecasting models were developed and compared using MAE, RMSE, and MAPE.

Model	MAE	RMSE	MAPE
SARIMA	18,031.40	19,009.18	18.97%
Prophet	20,250.79	22,318.41	21.86%
XGBoost	**14,763.81**	**18,337.41**	**14.48%**

XGBoost achieved the lowest forecasting errors and was selected as the production model for demand forecasting.

The model forecasts continued demand for major product categories over the next three months, with Furniture expected to achieve the highest sales, followed by Office Supplies and Technology.

Anomaly Detection Results

Anomaly detection was performed using the Video Game Sales dataset with two different techniques:

- Isolation Forest
- Z-Score Analysis

Isolation Forest identified the years **2007, 2008, 2009, and 2010** as anomalous because they experienced exceptionally high global sales compared with the overall dataset. These unusual sales levels likely resulted from major gaming console launches, blockbuster game releases, and increased market demand.

The Z-Score method detected fewer anomalies because yearly sales changed gradually rather than through sudden spikes.

Product Demand Segmentation

K-Means clustering grouped products into four demand segments:

Cluster	Description
High Value Premium Products	Copiers, Machines
Stable Demand Products	Appliances, Bookcases, Paper, Art, Labels, etc.
High Volume Best Sellers	Phones, Chairs, Tables, Storage, Accessories, Binders
Emerging Growth Product	Supplies

These segments help businesses prioritize inventory investment, improve warehouse management, and reduce stock shortages.

BUSINESS RECOMMENDATIONS

1. Prioritize Inventory for High-Demand Products

High Volume Best Sellers such as Phones, Chairs, Storage, and Accessories should receive priority inventory allocation to reduce stock-out risk and maximize sales.

2. Use XGBoost for Demand Forecasting

Since XGBoost achieved the highest forecasting accuracy, it should be adopted for routine demand forecasting and inventory planning.

3. Monitor Premium and Emerging Products

Premium products such as Copiers and Machines require careful inventory planning because of their high value. The Supplies category should be monitored closely because of its strong growth potential.

PROJECT LIMITATION

The forecasting models were trained using historical sales data only. External factors such as promotions, competitor activities, inflation, supply chain disruptions, and economic conditions were not included in the analysis. Incorporating these variables in future versions could further improve forecasting accuracy.

CONCLUSION

The developed Sales Forecasting and Demand Intelligence System demonstrates how machine learning and time-series forecasting can support better inventory planning and business decision-making. By combining sales forecasting, anomaly detection, demand segmentation, and an interactive Streamlit dashboard, the system provides actionable insights that help businesses improve operational efficiency, reduce inventory costs, and better meet customer demand.